

The Foundation for a Non Associative Closed Loop for Extraneous Solutions

Leonardo Urban

ORCID iD: 0009-0008-2147-026X

September 20, 2026

1 Core Axioms

The core axioms are as follows:

$$\sqrt{t} = -1 \tag{1}$$

$$t \neq \pm 1 \tag{2}$$

$$t^2 = 1 \tag{3}$$

2 Form of hyper complex number

A hyper complex number in this construction is currently represented by

$$z = a + bi + ct + dit \tag{4}$$

Where it is not a value but a hyper-complex unit.

3 Non-Associativity

Due to $t \neq \pm 1$ and $\sqrt{t} = -1$, Associativity breaks. This can be demonstrated by the following.

$$t = (\sqrt{t})^2 = (-1)^2 = 1 \tag{5}$$

Because of this we have to introduce non associativity, because of non associativity we need to define the following table.

$\cdot$	1	i	t	it
1	1	i	t	it
i	i	-1	it	t
t	t	$-it$	1	$-i$
it	it	$-t$	i	-1

The expression $i \cdot i \cdot t$ demonstrates the broken associativity:

$$(i \cdot i) \cdot t = (-1) \cdot t = -t \quad (6)$$

$$i \cdot (i \cdot t) = i \cdot it = t \quad (7)$$

Since $-t \neq t$, global associativity is no longer applicable, protecting the definition of $\sqrt{t} = -1$.

4 Attempting to define t^n

If we assume the definition of an exponent t^n as $e^{n \ln(t)}$ and let $e^{i\pi} = -1$ that means at $t^{1/2}$ or $\sqrt{t} = -1$ then $e^{i\pi} = t^{1/2}$ and $e^{i\pi} = e^{(1/2)\ln(t)}$ which makes $(1/2)\ln(t) = i\pi$ thus $\ln(t) = 2i\pi$. however $e^{2i\pi} = 1$ which would make $t = 1$ which breaks the second axiom.

4.1 Defining t^n where $n \in \mathbb{R}$

We need a system such that

n	real part of t^n	complex part of t^n
0	1	0
0.5	-1	0
1	0	1
1.5	0	-1
2	1	0

Where t^n repeats periodically to maintain the above table's properties. we can use $e^{i\theta} = \cos \theta + i \sin \theta$ as a placeholder for $t^{f(n)} = \cos(f(n)) + t \sin(f(n))$.

When n_{even} , $f(n_{\text{even}}) = \cos(0) + t \sin(0)$

i apologise whoever is writing this at the time im writing this midnight before this riemann bicentennial. i have the proofs but it boils down to

$$t^{f(n)} = \cos(f(n)) + t \sin(f(n)) \quad (8)$$

Where

$$f(n) = \pi \left(\frac{1}{2} \sin^2 \left(\frac{\pi n}{2} \right) + \frac{3}{4} \sin(n\pi) \right) \quad (9)$$

I am working to extend this definiton to all hypercomplex numbers but i am still very interested in hypercomplex spaces